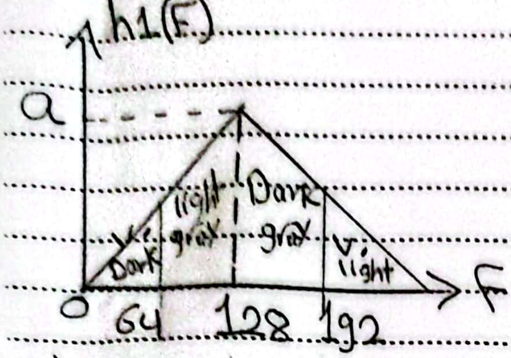
Fatma Mohamed 25 1000148



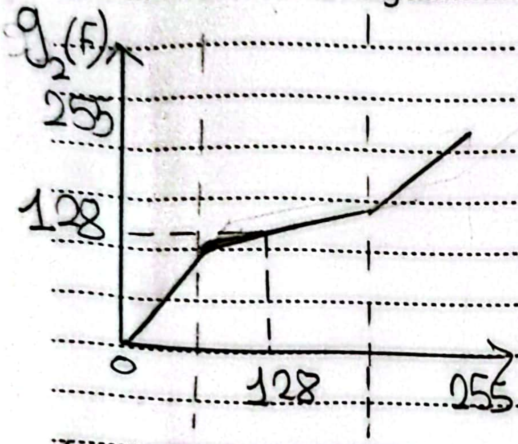
$h_1(F) \rightarrow g_2(F)$:-

* most of Pixels are grouped in in middle range (F=128).
* no. of very Dark, very Bright pixels are Low (zero, 255)

* to equalize it we need g_2:
- $(0 \rightarrow 64)$, $(192 \rightarrow 255)$:-

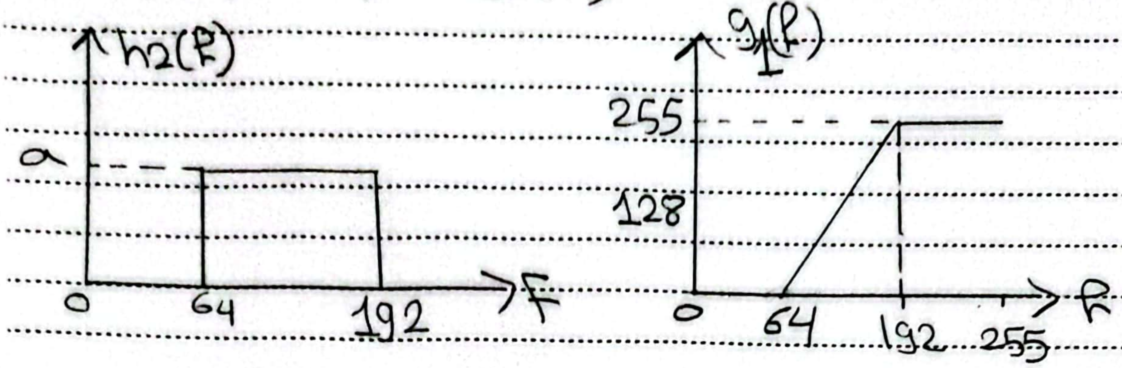
linear transformation function with high slope is applied to increase their frequencies.

- $(64 \rightarrow \sim 192)$ → transformation with low slope is applied to spread large no. of pixels clustred near (F = 128).



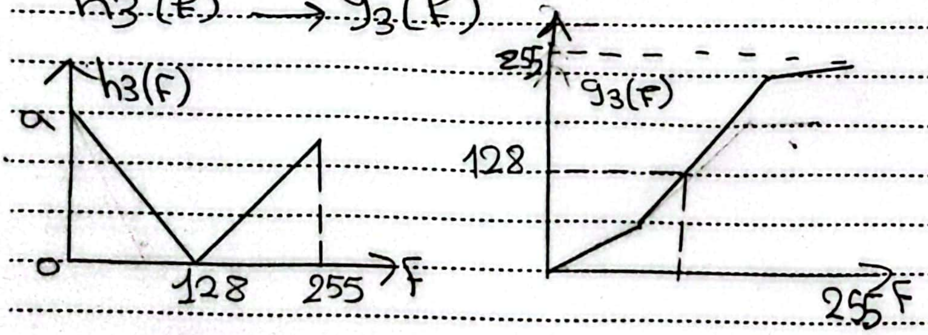
*(Low contrast Image)
(dull /grayish)

$h_2(F) \longrightarrow g_1(F)$



* all pixels in h2 between 64 → 192 and zero elsewhere so need linear mapping function that stretches it to [0 → 255].
* this achieved by g2 ⇒ below 64 → 0 and above 192 → 255

$h_3(F) \longrightarrow g_3(F)$



(very high Contrast Image)

* h3 has large no. of pixels at F=0, F=255 and low at F=128, so it needs g3 :- transformation with small slope is applied at F=0, F=255 to compress large no. of grouped pixels and slope is large at F=128 to stretch the small existing mid gray pixels.